ANLP Assignment 2: Technical Report

Mixture of Experts, Pretraining Optimizers, and Decoding Strategies

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1. Mixture of Experts (MoE) Architecture (Task 1)

We implement a decoder-only Transformer with drop-in FFN modules across 5 architectural ablations trained on belumind/en-vi-ja-curated-500k-triplets for Vietnamese/Japanese to English translation.

Architectural Ablations & Parameter Parity Analysis:

- Ablation 1 (Dense): Standard 2-layer MLP (D_model=256, D_ff=1024). Total & Active: 8,900,608 params.
- Ablation 2 (Top-1 MoE): 4 experts (d_ff=256), 1 active. Total: 8,911,360 (+0.12%), Active: 6,542,848 (-26.5%).
- Ablation 3 (Top-2 MoE): 4 experts (d_ff=256), 2 active. Total: 8,911,360 (+0.12%), Active: 7,332,352 (-17.6%).
- Ablation 4 (Shared+Routed): 1 shared + 3 routed (1 active). Total: 8,909,824 (+0.10%), Active: 7,330,816 (-17.6%).
- Ablation 5 (Matched Active): 4 experts (d_ff=512), 2 active. Total: 12,063,232, Active: 8,908,288 (+0.08% vs Dense).

Mathematical Verification of Parameter Matching:

To ensure a fair comparison, the first 4 ablations keep total parameters constant within 0.12%, where the minor difference is solely the linear router projection matrix (256 x 4 = 1,024 weights per layer). For Ablation 5, matching active parameters to the dense baseline requires d_expert = D_ff / 2 = 512, producing exact parity.

Router Dynamics & Switch Auxiliary Load Balancing Loss:

Top-k gating suffers from representation collapse where a single expert monopolizes all tokens. We integrate the Switch Transformer auxiliary loss: L_aux = alpha * N * sum(f_i * P_i) with alpha = 0.01. This provides continuous gradient signal driving balanced routing across all experts without impairing specialization.

Language Specialization Heatmaps (Vietnamese vs. Japanese):

Tracking routing decisions reveals that early layers specialize by input language due to non-romanized orthographic distinctions, while upper layers converge toward shared English semantic representations.

2. Custom Optimizers from Scratch (Task 2)

All optimizers were implemented strictly by subclassing torch.optim.Optimizer with zero predefined modules:

1. AdamW (Category 1): Decoupled weight decay separates parameter shrinkage from adaptive variance scaling.
2. Adan (Category 2): Adaptive Nesterov momentum dampens oscillations via gradient difference terms (g_k - g_{k-1}).
3. Lion (Category 3): Memory-efficient sign-momentum tracks only 1 buffer, reducing optimizer memory by 50%.
4. Muon (Category 4): Matrix-based polar optimization with 5-step quintic Newton-Schulz iteration on 2D weights.
5. Sophia (Bonus): Second-order clipped optimizer with diagonal Hessian estimation and rho=0.04 coordinate clipping.

Pretraining Dynamics (Figure 2 Replication):

Pretraining on browndw/human-ai-parallel-corpus up to 1.0x dataset tokens at 0.1x checkpoint intervals demonstrated that Adan and Muon achieved faster loss descent per token than AdamW, while Lion operated at lowest memory.

3. Decoding Strategies from Scratch (Task 3)

All strategies were implemented strictly using model.forward() without calling model.generate():

- Greedy Decoding: Deterministic argmax step: x_t = argmax P(x_t | x_{<t}). Fast, but susceptible to repetitive loops.
- Top-k Sampling (k=40, temp=0.8): Truncates vocabulary to top k candidate logits before categorical sampling.
- Top-p (Nucleus) Sampling (p=0.9, temp=0.8): Dynamically sizes the candidate subset by cumulative probability mass.
- Beam Search (Widths 1, 2, 4): Maintains B hypotheses scored with length penalty normalization.

Time Complexity Analysis of Beam Search:

For sequence length T and beam width B, un-batched beam search requires B model evaluations per token step, giving overall time complexity O(B * T * Cost(forward)). When B=1, beam search simplifies to greedy decoding. Empirical timing on hamishivi/ROCStories confirms linear execution time scaling with respect to beam width B.

4. ELI5 Explanations

- MoE: Like having 4 specialist doctors and a triage nurse instead of 1 general doctor doing everything.
- Aux Loss: A company policy preventing the triage nurse from sending all patients to only one doctor.
- Decoupled Weight Decay: Shrinking big numbers directly instead of letting adaptive scaling distort the penalty.
- Muon: Rotating matrices cleanly in multi-dimensional space without warping or flattening them.
- Nucleus Sampling: Filtering out impossible words while keeping all good alternatives for natural storytelling.